

## Necessary and Sufficient Conditions for Data-Driven Identification of Nonlinear Systems with Linear Embedding

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**Abstract:** This paper proposes data collection conditions for data-driven system identification of nonlinear systems that admit a Koopman linear embedding. Koopman operator-based system identification does not provide conditions on how data should be collected or how much data is required to guarantee identifiability. To address this issue, we integrate the identifiability condition from behavioral systems theory into a Koopman-based system identification framework. Since nonlinear systems that admit a linear embedding are generally not controllable, we adopt a relaxed version of the identification condition proposed by the fundamental lemma. This relaxed condition provides a criterion for data informativeness by imposing a rank condition on the Hankel matrix formed from the collected data. This ensures that the true system aligns with the most powerful unfalsified model, thereby guaranteeing identifiability. We extend this condition to nonlinear systems that admit a linear embedding, and establish a theoretical lower bound on the number of data points necessary for identifying nonlinear systems. Simulation results demonstrate that accurate system modeling and controller design are achievable with the minimal data satisfying this condition.

**Keywords:** Data-driven system identification, Koopman operator, Behavioral systems theory

### 1. INTRODUCTION

Accurately modeling dynamical systems is one of the fundamental problems in the field of control engineering. Although some systems have known mathematical models, many real-world systems are either too complex to model, are not accurately modeled, or change unpredictably over time. Consequently, data-driven approaches to control and system identification—based solely on input-output data—have gained significant attention in recent years.

The Koopman operator [1] enables a data-driven representation of nonlinear dynamical systems through a linear framework, when combined with dynamic mode decomposition (DMD) [2], [3]. However, when using the Koopman operator for system identification, the data collection process is often overlooked. In particular, it remains unclear how the data should be collected and how much data is required to identify a nonlinear system.

A partial answer to this problem can be found in the behavioral approach to system theory [4], [5]. The fundamental lemma from this framework shows that, under a persistently exciting input signal, the output of a system can be expressed as a linear combination of past trajectories. In other words, it provides conditions on how data should be collected and how much data is required to represent the system using data. However, this result is only applicable to linear time-invariant (LTI) systems and does not offer a method for explicitly modeling the system. As a consequence, it limits the applicability of a wide range of control policies.

Several studies have been conducted to overcome the limitations of these data-driven approaches. Data-enabled predictive control (DeePC) [6], [7] leverages the fundamental lemma to solve the model predictive control (MPC) problem using only previously measured trajec-

tories, without explicitly modeling the system. However, the paper demonstrates that DeePC is technically applicable to a very limited class of nonlinear systems, without providing any theoretical guarantees. In addition, as the prediction horizon increases, the number of past trajectory data points required also grows. Moreover, a key limitation is that control policies other than the MPC framework cannot be employed within this method.

Although there are still relatively few studies that combine the Koopman operator with the fundamental lemma, research in this direction has recently become more active. The work [8] enables data-driven prediction and control of nonlinear systems by combining the two aforementioned theories, but it utilizes the fundamental lemma solely for learning the Koopman lifting functions and does not provide any conditions for data collection. While some studies have employed the fundamental lemma to derive data collection conditions for Koopman-realized systems [9], [10], they do not carry out explicit system identification and are limited to the DeePC framework, thereby preventing the implementation of other control strategies.

To resolve these limitations, we adopt an identifiability condition from the behavioral systems theory and use it to establish an efficient data collection criterion for explicitly modeling nonlinear systems with linear embedding. The contributions of this study are as follows:

- By extending the rank condition on the Hankel matrix constructed from data to nonlinear systems, we establish a data collection condition that enables explicit modeling using the Koopman operator.
- We theoretically establish the minimum number of data points required for system identification in nonlinear systems that admit a Koopman linear embedding, by leveraging the depth of the Hankel matrix that satisfies the rank condition.

We numerically validate our method through simulations involving identification error evaluation and controller design. The simulation results demonstrate that, in accordance with our theoretical derivation, the proposed data collection condition enables successful explicit system identification. This is the first study that incorporates the identifiability condition of [11] into a Koopman-based system identification framework.

## 2. PRELIMINARIES

### 2.1 Dynamics

We consider a discrete-time nonlinear dynamical system evolving on a manifold  $\mathcal{X}$  such that,

$$\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k, \mathbf{u}_k), \quad \mathbf{y}_k = \mathbf{h}(\mathbf{x}_k, \mathbf{u}_k), \quad (1)$$

where  $\mathbf{x}_k \in \mathcal{X} \subseteq \mathbb{R}^{n_x}$ ,  $\mathbf{u}_k \in \mathcal{U} \subseteq \mathbb{R}^{n_u}$ ,  $\mathbf{y}_k \in \mathcal{Y} \subseteq \mathbb{R}^{n_y}$  denote the state, input, and output at time step  $k \in \mathbb{Z}$ , respectively.  $\mathbf{F}(\cdot, \cdot)$  defines the evolution operator, while  $\mathbf{h}(\cdot, \cdot)$  specifies the observation mapping.

### 2.2 Koopman Operator Theory

**Definition 1** (Koopman operator). *Consider a discrete-time dynamical system (1) with  $\mathbf{u}_k \equiv 0$ , i.e., the autonomous system. For the function of state space  $g \in \mathcal{F}$  with  $g : \mathcal{X} \rightarrow \mathbb{C}$ , which is called an observable, the Koopman operator  $\mathcal{K} : \mathcal{F} \rightarrow \mathcal{F}$  is defined by*

$$\mathcal{K}g(\mathbf{x}) = g \circ \mathbf{F}(\mathbf{x}),$$

where  $\mathcal{F}$  denotes a infinite-dimensional function space.

The Koopman operator is an infinite-dimensional, linear operator that evolves observables in the Hilbert space  $\mathcal{H}$ .

### 2.3 Extended dynamic mode decomposition

Due to the infinite-dimensional nature of the Koopman operator, finite-dimensional approximations are typically constructed via DMD [2], [12]. Among various DMD-based methods, extended dynamic mode decomposition (EDMD) [3] is one of the most widely used algorithms, due to its ability to effectively capture the system's nonlinearities through a rich set of observables, often referred to as lifting functions. Consider a dataset of snapshot pairs  $\{(\mathbf{x}_k, \mathbf{x}_k^+)\}_{k=1}^N$ , where  $\mathbf{x}_k^+ = \mathbf{F}(\mathbf{x}_k)$ . We assume that we can find the dictionary of observables  $\mathcal{D}_\psi = \{\psi_1, \psi_2, \dots, \psi_{N_K}\}$ , where  $\psi_i \in \mathcal{F}$ , which is rich enough to accurately approximate the Koopman operator. The linear span of  $\mathcal{D}_\psi$  is denoted by  $\mathcal{F}_{\mathcal{D}_\psi} \subset \mathcal{F}$ . We also define the vector-valued function  $\Psi : \mathcal{X} \rightarrow \mathbb{C}^{1 \times N_K}$ , where

$$\Psi(\mathbf{x}) = [\psi_1(\mathbf{x}) \quad \psi_2(\mathbf{x}) \quad \dots \quad \psi_{N_K}(\mathbf{x})].$$

The objective of EDMD is to construct a matrix  $\mathbf{K} \in \mathbb{C}^{N_K \times N_K}$ , a finite-dimensional approximation of the Koopman operator  $\mathcal{K}$ . By definition, a function  $\psi \in \mathcal{F}_{\mathcal{D}_\psi}$

can be written as

$$\psi(\mathbf{x}) = \sum_{k=1}^{N_K} a_k \psi_k(\mathbf{x}) = \Psi(\mathbf{x})\mathbf{a}.$$

Since  $\mathcal{F}_{\mathcal{D}_\psi}$  is not an invariant subspace of  $\mathcal{K}$  generally, there is a residual term,

$$(\mathcal{K}\psi)(\mathbf{x}) = (\Psi \circ \mathbf{F})(\mathbf{x})\mathbf{a} = \Psi(\mathbf{x})(\mathbf{K}\mathbf{a}) + r(\mathbf{x}).$$

To determine the matrix  $\mathbf{K}$ , we minimize the sum of residuals  $R$  defined as follows:

$$\begin{aligned} R &= \frac{1}{2} \sum_{k=1}^N |r(\mathbf{x}_k)|^2 \\ &= \frac{1}{2} \sum_{k=1}^N |(\Psi(\mathbf{x}_k^+) - \Psi(\mathbf{x}_k)\mathbf{K})\mathbf{a}|^2. \end{aligned}$$

The matrix  $\mathbf{K}$  that minimizes the residual  $R$  is given by the closed-form solution  $\mathbf{K} = \mathbf{G}^\dagger \mathbf{A}$ , where

$$\begin{aligned} \mathbf{G} &= \frac{1}{N} \sum_{k=1}^N \Psi(\mathbf{x}_k)^H \Psi(\mathbf{x}_k), \\ \mathbf{A} &= \frac{1}{N} \sum_{k=1}^N \Psi(\mathbf{x}_k)^H \Psi(\mathbf{x}_k^+). \end{aligned}$$

Here,  $(\cdot)^H$  and  $(\cdot)^\dagger$  denote the Hermitian adjoint and the Moore–Penrose pseudoinverse, respectively. Consequently, the infinite-dimensional observables in the function space are projected onto a finite-dimensional subspace  $\mathcal{F}_{\mathcal{D}_\psi}$ . This approach facilitates a data-driven approximation of the Koopman operator, thereby allowing for system identification.

### 2.4 Behavioral Systems Theory

The behavioral approach to system theory [4], [5] defines a system as a set of trajectories, and this perspective has recently attracted renewed attention in the context of data-driven representations. Following [13], we express a dynamical system and its properties in terms of its behavior [6].

**Definition 2** (Dynamical system [13]). *A dynamical system is defined as a 3-tuple  $\Sigma = (\mathbb{T}, \mathbb{W}, \mathcal{B})$ , where  $\mathbb{T} \subseteq \mathbb{Z}$  is the discrete-time axis,  $\mathbb{W}$  is a signal space, and  $\mathcal{B} \subseteq \mathbb{W}^{\mathbb{T}}$  is the behavior.*

Within the behavioral systems theory, analysis is typically restricted to LTI systems under the following assumptions.

**Assumption 1** (LTI system [6], [13]). *Let  $\Sigma = (\mathbb{T}, \mathbb{W}, \mathcal{B})$  be a dynamical system.*

- (i)  $\Sigma$  is linear.
- (ii)  $\Sigma$  is shift-invariant, i.e.,  $\mathcal{B} \subseteq \sigma\mathcal{B}$ , where  $\sigma$  denotes the forward time shift defined by  $(\sigma w)(t) = w(t + 1)$ .
- (iii)  $\Sigma$  is complete, i.e.,  $\mathcal{B}$  is closed in the topology of pointwise convergence.

This is equivalent to the following state-space representation:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k, \quad \mathbf{y}_k = \mathbf{C}\mathbf{x}_k + \mathbf{D}\mathbf{u}_k, \quad (2)$$

where the dimensions of the state, input and output variables are identical to those defined in (1).

### 2.5 Willems' Fundamental Lemma

In [13], Jan C. Willems demonstrated that the entire behavior space of a system can be expressed as a linear combination of trajectories that sufficiently excite the system's modes.

**Definition 3** (Persistently exciting). *Consider a sequence of vector signals*

$$\mathbf{f}_{[1:T]} \triangleq \{\mathbf{f}_n\}_{n=1}^T = [\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_T],$$

where  $\mathbf{f} : \mathbb{Z} \rightarrow \mathbb{R}^{\bullet}$ . The sequence  $\mathbf{f}_{[1:T]}$  is said to be persistently exciting of order  $L$  if its Hankel matrix of depth  $L$

$$\mathcal{H}_L(\mathbf{f}_{[1:T]}) = \begin{bmatrix} \mathbf{f}_1 & \mathbf{f}_2 & \cdots & \mathbf{f}_{T-L+1} \\ \mathbf{f}_2 & \mathbf{f}_3 & \cdots & \mathbf{f}_{T-L+2} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{f}_L & \mathbf{f}_{L+1} & \cdots & \mathbf{f}_T \end{bmatrix}$$

has full row rank.

**Lemma 1** (Willems' fundamental lemma). *Consider a behavior  $\mathcal{B}$  under Assumption 1, whose input-state-output trajectory tuple is  $(\mathbf{u}_{[0:T-1]}, \mathbf{x}_{[0:T-1]}, \mathbf{y}_{[0:T-1]})$ . Assume that  $\mathcal{B}$  is controllable and the input sequence  $\mathbf{u}_{[0:T-1]}$  is persistently exciting of order  $n_x + L$ . Then, the following properties are satisfied.*

(i) *The Hankel matrix*

$$\mathcal{H}_0 \triangleq \begin{bmatrix} \mathcal{H}_1(\mathbf{x}_{[0:T-L]}) \\ \mathcal{H}_L(\mathbf{u}_{[0:T-1]}) \end{bmatrix} = \begin{bmatrix} \mathbf{x}_0 & \mathbf{x}_1 & \cdots & \mathbf{x}_{T-L} \\ \mathbf{u}_0 & \mathbf{u}_1 & \cdots & \mathbf{u}_{T-L} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_{L-1} & \mathbf{u}_L & \cdots & \mathbf{u}_{T-1} \end{bmatrix}$$

has full row rank. [13, Corollary 2]

(ii) *Any input-output sequence  $[\tilde{\mathbf{u}}^\top, \tilde{\mathbf{y}}^\top]^\top \in \mathbb{R}^{(n_u+n_y)L}$  is a valid trajectory of (2) if and only if there exists  $\mathbf{v} \in \mathbb{R}^{T-L+1}$  such that*

$$\begin{bmatrix} \mathcal{H}_L(\mathbf{u}_{[0:T-1]}) \\ \mathcal{H}_L(\mathbf{y}_{[0:T-1]}) \end{bmatrix} \mathbf{v} = \begin{bmatrix} \tilde{\mathbf{u}} \\ \tilde{\mathbf{y}} \end{bmatrix}.$$

[13, Theorem 1]

Lemma 1 implies that, under the assumption that  $\mathbf{u}_{[0:T-1]}$  is persistently exciting, the future outputs of the LTI system can be fully reconstructed from the given inputs as a linear combination of past trajectories.

## 3. DATA-DRIVEN IDENTIFIABILITY

### 3.1 Notations

We begin by introducing the terminology used throughout the paper. Let  $\mathcal{W}$  denote the entire dataset, and let  $\mathcal{W}_d \subseteq \mathcal{W}$  represent the subset of observed data used for system identification, whose element is  $w_d$ . We denote by  $\mathcal{M}$  the model class, which contains candidate representations of the system behavior. The true system—that is, the one that actually generated the data—is denoted by  $\bar{\mathcal{B}}$ , and  $\mathcal{B}|_L$  denotes the set of truncated behaviors consisting of only the first  $L$  time steps of its trajectory. The lag  $l$  (equivalent to the observability index) is defined by the smallest integer for which the observability matrix  $\mathcal{O}_l(\mathbf{A}, \mathbf{C}) := \text{col}(\mathbf{C}, \mathbf{C}\mathbf{A}, \dots, \mathbf{C}\mathbf{A}^{l-1})$  has full rank, i.e.,  $\text{rank}(\mathcal{O}_l) = n_x$ . We define the model complexity of an LTI system  $\mathcal{B}$  by the triplet  $\mathbf{c}(\mathcal{B}) := (n_u, l, n_x)$  [11], and assume that the elements of the model complexity triplet are known a priori.

### 3.2 Identifiability of controllable LTI systems

First, we define the identifiability in a behavioral systems theory.

**Definition 4** (Identifiability [11]). *A dynamical system  $\bar{\mathcal{B}} \in \mathcal{M}$  is identifiable from the data  $\mathcal{W}_d \subset \bar{\mathcal{B}}$  in the model class  $\mathcal{M}$  if  $\bar{\mathcal{B}}$  is the only model in  $\mathcal{M}$  that fits the data exactly, i.e.,*

$$\hat{\mathcal{B}} \in \mathcal{M} \quad \text{and} \quad \mathcal{W}_d \subset \hat{\mathcal{B}} \quad \Rightarrow \quad \hat{\mathcal{B}} = \bar{\mathcal{B}}.$$

Lemma 1 provides the following identifiability condition.

**Theorem 1** (Sufficient conditions for identifiability of an LTI system [13]). *An LTI system  $\mathcal{B}$  is identifiable if*

- (i)  *$\mathcal{B}$  is controllable;*
- (ii)  *$\mathcal{W}_d$  can be partitioned into input and output components;*
- (iii)  *$u_d$  is persistently exciting of order  $l + n_x + 1$ , i.e., the rank  $\mathcal{H}_{l+n_x+1}(u_d) = n_u(l + n_x + 1)$ .*

The proof can be found in [11].

### 3.3 Koopman linear embedding

To extend the identifiability condition to nonlinear systems, we leverage the Koopman operator framework to transform the original nonlinear system into a lifted linear representation. To ensure theoretical completeness, we restrict our attention to nonlinear systems for which there exists a set of lifting functions that yield an exact linear representation. Furthermore, we assume that such a set of lifting functions can always be found.

**Definition 5** (Koopman linear embedding [9]). *The nonlinear system (1) admits a Koopman linear embedding if there exists a basis vector of lifting functions*

$$\mathbf{z}_k := \Phi(\mathbf{x}_k) = [\phi_1(\mathbf{x}_k), \phi_2(\mathbf{x}_k), \dots, \phi_{n_z}(\mathbf{x}_k)]^\top \in \mathbb{R}^{n_z}$$

such that it evolves linearly along all trajectories of (1).

The class of nonlinear systems that admit the Koopman linear embedding, as defined above, satisfy our assumptions. Assuming that the system is fully observable, we can identify both the system matrix and the input matrix using EDMD, given  $N$  snapshot data samples,

$$(\mathbf{A}, \mathbf{B}) \in \arg \min_{\mathbf{A}, \mathbf{B}} \|\mathbf{Z}^+ - \mathbf{AZ} - \mathbf{BU}\|_F^2,$$

where  $\mathbf{Z} = [\Phi(\mathbf{x}_0), \dots, \Phi(\mathbf{x}_{N-2})]$ ,  $\mathbf{Z}^+ = [\Phi(\mathbf{x}_1), \dots, \Phi(\mathbf{x}_{N-1})]$ ,  $\mathbf{U} = [\mathbf{u}_0, \dots, \mathbf{u}_{N-2}]$  and  $\|\cdot\|_F$  denotes the Frobenius norm. While this approach enables the representation of a nonlinear system in the form of an LTI system, the pair  $(\mathbf{A}, \mathbf{B})$  is, in general, not controllable. Therefore, the identifiability condition provided by the fundamental lemma is not sufficient in this case, and an alternative condition that relaxes this requirement is needed.

### 3.4 Partial extension to nonlinear systems

In this subsection, we apply the identifiability condition proposed in [11] to nonlinear systems that admit a Koopman linear embedding. Based on this condition, we derive the minimum number of data samples required for the identification of such nonlinear systems. We introduce the concept of the most powerful unfalsified model (MPUM), which is essential for understanding a more general identifiability condition beyond that of Theorem 1.

**Definition 6** (The most powerful unfalsified model [14]). *If a system  $\mathcal{B}$  is the most powerful unfalsified model  $\mathcal{B}_{\text{MPUM}}$  for the data  $\mathcal{W}_d$  in the model class  $\mathcal{M}$ , the following properties are satisfied.*

- (i)  $\mathcal{B}$  is in the model class, i.e.,  $\mathcal{B} \in \mathcal{M}$ .
- (ii)  $\mathcal{B}$  is unfalsified, i.e.,  $\mathcal{W}_d \subseteq \mathcal{B}$ .
- (iii)  $\mathcal{B}$  is most powerful, i.e., any other model  $\mathcal{B}'$  for which properties (i) and (ii) hold is no more powerful than  $\mathcal{B}$ .

$$\mathbf{c}(\mathcal{B}) \leq \mathbf{c}(\mathcal{B}') \quad (\mathcal{B} \subseteq \mathcal{B}')$$

The  $\mathcal{B}_{\text{MPUM}}$  refers to the simplest model that remains consistent with the observed data, i.e.,  $\mathcal{B}_{\text{MPUM}}$  is the solution of optimization problem  $\min_{\mathcal{B} \in \mathcal{M}, \mathcal{W}_d \subseteq \mathcal{B}} \mathbf{c}(\mathcal{B})$ . The existence and uniqueness of the most powerful unfalsified model are guaranteed, and based on this fact, the following identifiability condition is derived.

**Theorem 2** (Necessary and sufficient conditions for identifiability of an LTI system [11]). *Let the data  $\mathcal{W}_d$  be generated by a system  $\mathcal{B}$  whose input dimension, state dimension and lag are  $n_u$ ,  $n_x$ ,  $l$ , respectively. Then  $\mathcal{B}$  is identifiable from  $\mathcal{W}_d$ , i.e.,  $\mathcal{B}_{\text{MPUM}}(\mathcal{W}_d) = \mathcal{B}$  if and only if*

$$\text{rank } \mathcal{H}_{l+1}(\mathcal{W}_d) = n_u(l+1) + n_x.$$

For a detailed proof, the reader is referred to [11]. This condition does not require controllability, and is therefore applicable to nonlinear systems that admit the Koopman linear embedding. Moreover, since it does not require input/output partitioning, the condition is also applicable

to autonomous systems. Based on this condition, we can now derive the necessary and sufficient conditions of the data and the minimum number of data samples required for identifying nonlinear systems which admit the Koopman linear embedding.

**Corollary 1** (Necessary and sufficient conditions for identifiability of a nonlinear system with linear embedding). *Assume the nonlinear system (1) admits a Koopman linear embedding. Then the system (1) is identifiable if and only if  $\text{rank } \mathcal{H}_{l+1}(\mathcal{W}_d|_T) = n_u(l+1) + n_z$  holds for  $T \geq (n_u + 1)(l+1) + n_z - 1$ .*

**Proof:** The condition,  $\text{rank } \mathcal{H}_{l+1}(\mathcal{W}_d|_T) = n_u(l+1) + n_z$ , is derived from Theorem 2, where  $n_z$  denotes the dimension of the lifted state. From the rank condition of the Hankel matrix, we observe that the number of its rows,  $T-l$ , must be greater than or equal to  $n_u(l+1) + n_z$ , i.e.,  $\text{rank } \mathcal{H}_{l+1}(\mathcal{W}_d|_T)$ . Therefore, it follows that  $T-l \geq n_u(l+1) + n_z$ , which completes the proof. ■

By the same logic, for a general nonlinear system with a residual sum  $R$ , we can define a strict lower bound on the number of data samples required for system identification.

$$T > (n_u + 1)(l+1) + n_z - 1.$$

## 4. NUMERICAL VALIDATIONS

In this section, we present numerical validations of the data-driven system identification framework within simulations. The simulation is conducted in MATLAB® environment, under the desktop configuration of Intel® Core™ i7-14700 CPU and 32 GB of RAM. We consider the nonlinear system that admits Koopman linear embedding, i.e., without residual term  $R$ . We use the benchmark example of the nonlinear system in [9],

$$\begin{bmatrix} x_{1,k+1} \\ x_{2,k+1} \end{bmatrix} = \begin{bmatrix} 0.99 x_{1,k} + 1.1 x_{1,k}^3 + x_{1,k}^4 + 0.5 u_k \\ 0.9 x_{2,k} + 1.2 x_{1,k}^2 + x_{1,k}^3 + x_{1,k}^4 + 0.5 u_k \end{bmatrix}, \quad (3)$$

with output  $\mathbf{y}_k = \mathbf{x}_k$ , and state  $\mathbf{x} = [x_1, x_2]^\top \in \mathbb{R}^2$ . We choose  $\mathbf{z} := [x_1, x_2, x_1^2, x_1^3, x_1^4]^\top$  as the lifted state, and its Koopman linear embedding becomes

$$\mathbf{z}_k = \begin{bmatrix} 0.99 & 0 & 0 & 0 & 0 \\ 0 & 0.9 & 1.2 & 1.1 & 1 \\ 0 & 0 & 0.99^2 & 0 & 0 \\ 0 & 0 & 0 & 0.99^3 & 0 \\ 0 & 0 & 0 & 0 & 0.99^4 \end{bmatrix} \mathbf{z}_k + \begin{bmatrix} 0 \\ 0.5 \\ 0 \\ 0 \\ 0 \end{bmatrix} u_k.$$

The system is not controllable, and has an observability index 4. According to Corollary 1, data-driven system identification is guaranteed if at least

$$T_{\min} = (n_u + 1)(l+1) + n_z - 1 = 14 \quad (4)$$

trajectory data points are collected, and the rank condition

$$\text{rank } \mathcal{H}_{l+1} = n_u(l+1) + n_z = 10 \quad (5)$$

is achieved. To satisfy conditions (4) and (5), we generated 14 random input signals uniformly drawn from a



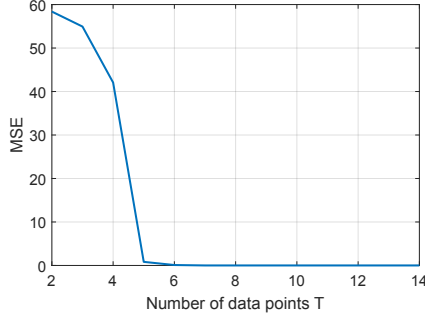


Fig. 1.: Mean square error of the system (3)’s output signals with respect to the number of data points  $T$ .

normal distribution with a standard deviation of 1. After confirming that the rank condition of the Hankel matrix constructed from the data  $w_d = [u, \mathbf{y}^\top]^\top$  is satisfied with rank 10, we proceed with system identification using EDMD.

Fig. 1 shows that the identification error decreases as the number of data points approaches  $T_{\min}$ . The identification error is defined as the mean squared error (MSE) given by:

$$\text{MSE} = \frac{1}{N} \sum_{k=1}^N \|\mathbf{y}_k - \hat{\mathbf{y}}_k\|_2^2,$$

where  $\mathbf{y}_k$  denotes the output from the true system, and  $\hat{\mathbf{y}}_k$  denotes the output generated by the identified system at time step  $k$ . We set the number of data snapshots to  $N = 200$ , and evaluated the MSE using a different random input signal than the one used during system identification. It can be observed that as the number of data points used for system identification approaches  $T_{\min} = 14$ , the MSE converges to zero. Although Fig. 1 shows that the MSE becomes nearly zero for  $T \geq 6$ , suggesting that the system has been successfully identified in this specific case, Corollary 1 provides a theoretical guarantee of identifiability only when  $T \geq T_{\min} = 14$  in the general case.

Fig. 2 demonstrates that the linear quadratic regulator (LQR), designed using the model identified from 14 data points, performs appropriately on the nonlinear system. The LQR controller is designed to minimize the following cost function:

$$\hat{J} = \sum_{k=0}^{\infty} [\mathbf{z}_k^\top \hat{\mathbf{Q}} \mathbf{z}_k + u_k^\top \mathbf{R} u_k],$$

where

$$\hat{\mathbf{Q}} = \begin{bmatrix} \mathbf{Q} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{n_z \times n_z}, \quad \mathbf{Q} = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}, \quad \mathbf{R} = 0.01.$$

An affine feedback controller

$$u(k) = -K_{\text{LQR}} \mathbf{z}_k + u_0$$

solves the optimal control problem where  $K_{\text{LQR}}$  and  $u_0$  denotes the LQR gain and the drift term, respectively. The

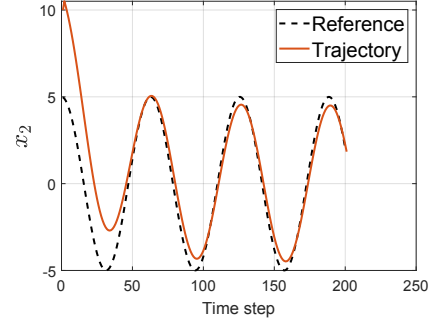


Fig. 2.: Trajectory of  $x_2$  over time steps under the LQR controller designed based on the identified model.

reference trajectory is defined as  $r(t) = 5 \cos(0.1t)$  and the initial state is  $\mathbf{x}_0 = [10, 10]^\top$ . The control input generated by the LQR was applied to the original nonlinear system. We confirm that the model-based controller, designed using a model identified from only 14 data points, successfully achieves the control goal.

## 5. CONCLUSIONS

In this paper, we derived data collection conditions for nonlinear system identification, which admits Koopman linear embedding. We introduced an identifiability condition from behavioral systems theory that does not require the controllability assumption, which shows that an LTI system is identifiable if the rank condition of a Hankel matrix of a certain depth is satisfied. From this, we theoretically derived the minimum number of data points required for the nonlinear system identification with linear embedding. Numerical simulations demonstrated that the nonlinear system, identified using only the theoretically minimal data, accurately captures the system dynamics. Our future work will focus on characterizing data conditions that ensure identifiability, even when the Koopman-based model of a nonlinear system yields nonzero but bounded residuals.

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